Benedict Armstrong

Curriculum Vitae

Machine learning for physics · Computational Science and Engineering, ETH Zürich
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Computational Science and Engineering student at ETH Zürich working at the intersection of machine learning and physics. My research focuses on sequence models for detecting and characterising gravitational-wave signals. I also have several years of experience shipping full-stack software as a freelancer and have taught algorithms, data structures and software engineering to hundreds of students.

1 Education



Figure 1: Benedict Armstrong, Zürich. Born and raised in Vienna, currently living in Zürich. In my free time, I like to code, tinker with electronics, design and make furniture, and generally enjoy all types of DIY/maker projects. I am also interested in FPV drones, IoT prototyping, 3D printing, tennis, cycling, triathlons, and of course, all things computer science.

M.Sc. Computational Science and Engineering
ETH Zürich — 2026 – present
Robotics track

B.Sc. Computational Science and Engineering
ETH Zürich — 2021 – 2025
Robotics track

B.Sc. Computer Science
Vienna University of Technology — 2019 – 2021 (switch to ETH)
Software & Information Engineering track

2 Research

AI for Gravitational-Wave Physics
ML Research Intern, Computational Applied Mathematics & AI Lab (CAMAIL) — June 2025 – Summer 2026
ELLIS Institute Tübingen and Max Planck Institute for Intelligent Systems, in collaboration with the LIGO Lab at MIT supervised by T. Konstantin Rusch.

When two neutron stars spiral into each other, they send ripples through spacetime that LIGO measures as length changes far smaller than a proton. These signals can last minutes and are buried in non-stationary, glitchy detector noise, while astronomers need an alert within seconds to point telescopes at the merger. I work on Linear Oscillatory State-Space Models (LinOSS): sequence models that handle very long inputs efficiently and make it possible to detect binary neutron star mergers in real time.

- Designed and trained state-space models for low-latency binary neutron star detection on LIGO strain data.
- Built the training and evaluation pipeline in PyTorch (Lightning) and JAX/Equinox.
- Co-developed PhyTS, a benchmark of physics time-series tasks (NeurIPS 2026).

Results appeared at NeurIPS 2026 [1] and were presented at AI for Gravitational Waves Workshop (CERN, Geneva) [2] and Fast Machine Learning for Science Conference 2026 (UC San Diego) [3]. *Tools:* Python, PyTorch, PyTorch Lightning, JAX, Equinox.

3 Publications and Talks

[1] **Benedict Armstrong**, Jeroen Audenaert, Hannah P. Binney, Alice Cheng, Maarten De Vos, Luis F. Domingues, et al. (26 authors). PhyTS: A Benchmark for Scientific Time Series. In *Advances in Neural Information Processing Systems (NeurIPS)*, Sydney, 2026.

[2] Benedict Armstrong. Linear Oscillatory State-Space Models for Binary Neutron Star Detection. Talk at *AI for Gravitational Waves Workshop*, CERN, Geneva, May 2026.

[3] Kevin (Kyungseop) Yoon. Direct Parameter Estimation of Binary Neutron Star Mergers with State Space Models. Talk at *Fast Machine Learning for Science Conference 2026*, UC San Diego, September 2026.

4 Experience

Freelance Software Engineer
Armstrong Systems 2022 – 2024
Founded Armstrong Systems, software consulting for clients in Zürich. Full-stack development of various projects including a web-based data management tool, visualization dashboards and large-scale data ingestion pipelines.

- Scoped, built and maintained a web-based data management tool used day to day by the client's team.
- Built interactive visualization dashboards on top of the clients' operational data.
- Designed large-scale data ingestion pipelines, from source integration to deployment and monitoring.

Coding Assistant for LehrFEM++ Finite Element Library
Seminar for Applied Mathematics, ETH Zürich Fall 2024
Coding assistant for the C++ finite element library LehrFEM++, used in the course *Numerical Methods for Partial Differential Equations* taught by Prof. Dr. Ralf Hiptmair.

- Extended and maintained the LehrFEM++ codebase used by students in the course's programming exercises.
- Wrote documentation and examples to make the library easier for students to pick up.

Paramedic (civil service)
Samariterbund 2018 – 2019
Worked as a paramedic in Vienna as part of my civil service.

Various summer jobs
Various Summers 2017–2021
Various summer jobs in Vienna, including working as a postman, food delivery driver and math/CS tutor for high school students.

5 Teaching

Table 1: Teaching positions, grouped by institution.

Role	Course	Period
Department of Computer Science, ETH Zürich		
Teaching Assistant	*Software Engineering Fundamentals*	Spring Semester 2024
Head Teaching Assistant	*Computer Science II*	August 2023 – February 2024
Teaching Assistant	*Computer Science II*	August 2022 – February 2023
Research Group Industrial Software (INSO), TU Wien		
Teaching Assistant and Full Stack Developer	*Security for Systems Engineering*	Spring 2021

6 Projects

Presio · presio.ch
Web app for presenting PDF slides with a presenter controller and a mirrored audience window. PDFs stay in the browser by default; signed-in users can share a session code so others can follow live on their own devices.

AcceptOdds · acceptodds.com
Short description of what AcceptOdds does, who it is for and what is technically interesting about it.

7 Skills

Table 2: Technical skills.

Area	Tools
Machine learning	Python, PyTorch, JAX
Languages	C++, C, Java, JavaScript, PHP, SQL
Web	Angular, React, NextJS, Svelte(Kit), FastAPI, REST, HTML, CSS
Data & infrastructure	Mongo, Postgres, Docker, Git
Other	Arduino/ESP8266/32 based hardware development, Scrum
Spoken languages	English, German